
From DDPM to Rectified Flow: a Measured, From-Scratch Diffusion Study

Yi Hou

University of Chinese Academy of Sciences
Beijing, China
houyi25@mailsucas.ac.cn

Abstract

This report documents a from-scratch implementation and measurement study of the modern diffusion stack. No `diffusers`, `flash_attn` or pretrained generative component is used: DDPM, DDIM, a diffusion transformer (DiT) with adaLN-Zero conditioning, and rectified flow are all implemented in this repository, together with the training pipeline, the sampler suite and the evaluation code. The study is organized as a controlled 2×2 design (UNet vs. DiT $\times$ ϵ -prediction vs. rectified flow) at two scales, MNIST 28×28 and CelebA 64×64 , with every cell trained under one frozen protocol. Headline results: DDIM at 50 function evaluations recovers the 1000-step ancestral FID to within 12% on CelebA (12.75 vs. 11.37) at $5.2 \times$ the sampling throughput, and rectified flow dominates the quality–NFE frontier on MNIST (FID 0.88 at 64 evaluations, versus 4.32 for the best ϵ -prediction sampler at 50). The same deterministic sampler that improves the UNet collapses on the transformer — its FID grows from 49 to 672 as NFE goes $10 \rightarrow 50$, against 33 for 1000-step ancestral sampling of the same weights — and an η sweep shows the failure is the missing noise injection. On the systems side, `bf16` buys $3.15 \times$ step time and $1.6 \times$ memory over `fp32` while `torch.compile` buys nothing ($1.00 \times$), and a kernel-level profile attributes 67% of GPU time to matmuls and 16% to elementwise work, which is where a fused adaLN kernel would have to earn its keep.

1 Introduction

Diffusion models are usually consumed through a library: a scheduler object, a pretrained backbone, a pipeline. This report takes the opposite route. Everything is built from scratch — the forward and reverse processes, the two backbones, both training objectives, the samplers and the evaluation — and every claim is backed by a measurement produced by this repository.

The study asks three questions:

- **RQ1 (architecture).** At a similar parameter scale, does replacing a UNet with a transformer (DiT) backbone help or hurt generation quality and computational efficiency?
- **RQ2 (objective and sampler).** How does a rectified-flow objective change the quality–NFE trade-off, and how much does sampler choice alone buy for a fixed trained model (DDPM ancestral vs. DDIM)?
- **RQ3 (systems).** How much training and inference throughput can be recovered by mixed precision, compilation and kernel-level fusion at this scale — and where does a distributed strategy stop being worth it?

Table 1: Results at a glance. All numbers are measured on the hardware described in Section 4; the raw archives live in `results/*.json`.

Question	Baseline	Measured effect
RQ1 backbone	UNet (11.7M params, 2.43 GFLOPs/sample)	DiT (16.5M, 1.08): worse FID at every matched budget (MNIST ϵ : 2.79 vs. 33.2; flow: 0.88 vs. 4.99) despite $2.25\times$ fewer FLOPs
RQ2 objective	ϵ -prediction	rectified flow: FID 0.88 vs. 4.32 at NFE 64 vs. 50; crossover near NFE 16
RQ2 sampler	DDPM ancestral (1000 NFE)	DDIM at 50 NFE: FID 12.75 vs. 11.37 on CelebA ($5.2\times$ throughput); collapses on the DiT (49 $\rightarrow$ 672 as NFE 10 $\rightarrow$ 50)
RQ3 systems	FP32, uncompiled	BF16 $3.15\times$ step time, peak memory 32.9 $\rightarrow$ 21.1 GB, MFU 10.4% $\rightarrow$ 32.7%; <code>compile</code> 1.00 $\times$
RQ3 profile	—	matmul 67.1%, elementwise 15.6%, attention 11.1%; the fused-adaLN target is the elementwise slice

Table 2: The three objectives on a common footing. $\bar{\alpha}_t$ is the usual DDPM cumulative product; the flow variant uses a continuous $t \in [0, 1]$.

Method	Path x_t	Target	Sampler (NFE)
DDPM [1]	$\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$	ϵ	ancestral, stochastic (1000)
DDIM [2]	same	ϵ	deterministic ODE (10–50)
Rectified flow [8, 9]	$(1 - t) x_0 + t \epsilon$	$v = \epsilon - x_0$	Euler ODE (4–128)

Section 2 puts the three objectives on a common footing. Section 3 describes the implementation, Section 4 the frozen protocol and the evaluation protocol, Sections 5–7 the results, and Section 8 collects the experiments that failed, which were as instructive as the ones that worked.

2 Background: one path, three objectives

All three methods place the data and the noise at the ends of an interpolation path and train a network to regress a target along it. They differ in the path, in the target, and in the solver used to walk back from noise to data.

One implementation detail is easy to get wrong and worth stating: the sinusoidal timestep embedding used by both backbones is calibrated for $t \in [0, 1000]$ (the DDPM range). The flow code therefore keeps its path mathematics in $t \in [0, 1]$ but passes $t \times 1000$ to the network; without that scaling the embedding is nearly constant and conditioning collapses.

3 Implementation

Everything is written in PyTorch, with no generative-model library. Table 3 lists the measured sizes. Note that the parameter ranking and the FLOPs ranking disagree across scales — on CelebA the UNet is larger on both axes, on MNIST it has fewer parameters but more FLOPs — which is exactly why both are reported.

Backbones. The UNet is a residual U-net with GroupNorm, SiLU, sinusoidal timestep injection into every residual block, and `scaled_dot_product_attention` at the bottleneck. The DiT follows (author?) [7]: patchify with a strided convolution, fixed two-dimensional sincos positional embeddings, and blocks whose attention and MLP are modulated by an adaLN-Zero MLP that is zero-initialized, so every block starts as the identity (verified in `tests/test_sanity.py`: the untrained model outputs exactly zero). Attention uses PyTorch SDPA in both backbones; no fused-attention library is used.

Training. A single training loop drives both objectives. It supports `torchrun` DDP, YAML configs, checkpoint resume, an exponential moving average of the weights (all reported FID uses EMA weights), per-epoch EMA-only snapshots for quality-vs-progress curves, and cost account-

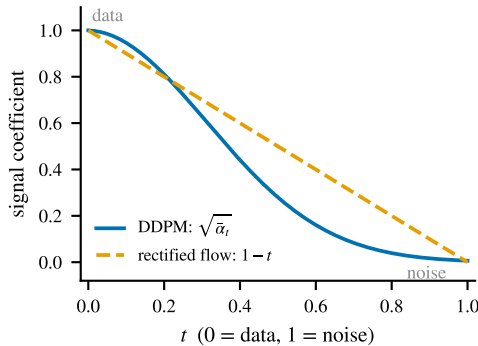


Figure 1: The diffusion path and the rectified-flow path between data and noise. The flow objective regresses a constant velocity along a straight line; the diffusion objective traverses a curved, non-uniform-speed path.

Table 3: Measured sizes (forward pass, one sample at the model’s native resolution). Params and FLOPs are counted by `eval/cost.py` with `torch.utils.flop_counter`, so both backbones are measured with the same tool.

Dataset	Backbone	Params	GFLOPs/sample	Conditioning	Attention
MNIST 28 ²	UNet	11.7M	2.43	sinusoidal + MLP	bottleneck
MNIST 28 ²	DiT	16.5M	1.05	adaLN-Zero	all 6 blocks (49 tokens)
CelebA 64 ²	UNet	174.1M	67.8	sinusoidal + MLP	bottleneck
CelebA 64 ²	DiT	129.6M	43.6	adaLN-Zero	all 12 blocks (256 tokens)

ing: parameters, measured FLOPs, step time, throughput, peak memory and MFU are written to `results/<run>.json` on every run.

Samplers. Ancestral DDPM sampling, deterministic DDIM sampling with a configurable NFE, and Euler integration of the flow ODE. All three reuse the same trained ϵ - or velocity-predicting network, so they can be compared at fixed weights.

4 Experimental setup

Hardware. TODO: fill in the machine actually used (GPU model, interconnect, NUMA layout). The topology matters: on the six-A6000 node used for the systems numbers, GPUs 0–3 and 4–5 sit in different NUMA domains and the cross-domain link is an order of magnitude slower, which is why `NCCL_P2P_DISABLE=1` is required and why DDP is kept inside one domain.

Protocol. All cells of a scale share one frozen protocol: identical optimizer, learning-rate schedule, batch size (global), EMA decay and seed (Table 4). Unlike recipes that tune each cell separately, this design means a difference between cells is attributable to the backbone or the objective rather than to the optimizer.

FID protocol. Reference statistics are computed from the training tensor itself (`data/celeba_64_uint8.pt` for CelebA, the MNIST training split for MNIST) with the same preprocessing chain used to evaluate samples; third-party statistics are never used, because any difference in resizing makes the numbers incomparable. CelebA uses the standard Inception-v3 pool3 features at 299². Inception features are not meaningful for grayscale digits, so the MNIST cells are evaluated in the feature space of a small convolutional classifier trained on MNIST inside this repository (Table 4 footnote); that metric is *non-standard* and only comparable within this report. Tier 1 evaluations use 10k samples, Tier 2 (final) 50k, and the spread is reported as the variability across five random 10k subsets.

Table 4: The frozen protocol. Any deviation is recorded in the run’s JSON.

	MNIST 28 ²	CelebA 64 ²
Cells	UNet/DiT $\times \epsilon$ /flow, one run each	
Epochs	200	300
Global batch	256	800 (160 per GPU $\times$ 5)
Optimizer	Adam, lr 2×10^{-4} , warmup then cosine to 10^{-6}	
Gradient clipping	1.0	
EMA decay	0.9999, applied in every cell	
Seed	42	

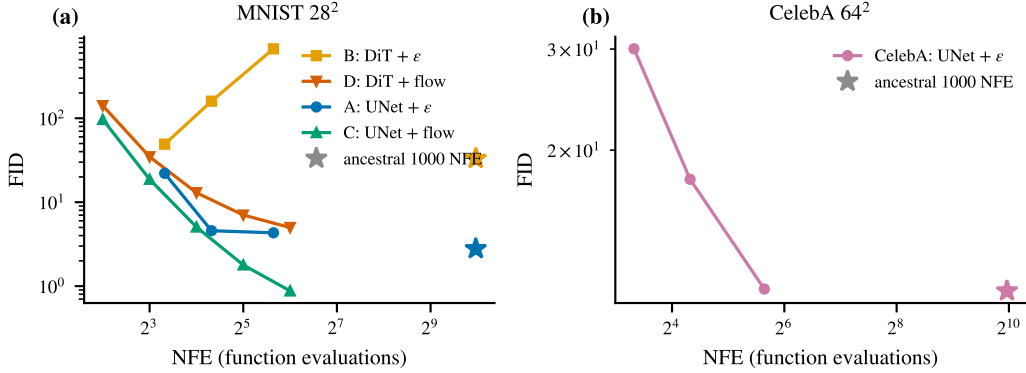


Figure 2: Quality against the number of function evaluations, by dataset (note the logarithmic axes and the separate panels: the CelebA FIDs sit an order of magnitude above the MNIST ones). Within a panel every curve is a trained model plus a sampler; the DDPM vs. DDIM pair is a same-weights sampler ablation, while comparisons across cells also change the training objective.

5 Sampler study (RQ2)

5.1 What a better sampler buys, at CelebA scale

The practical promise of DDIM is that a fixed trained model can be sampled in far fewer network evaluations. Table 5 answers that for the existing CelebA DDPM checkpoint (UNet, 174.1M parameters, 300 epochs, trained before this protocol freeze). Quality improves monotonically with the budget, and the 50-step deterministic sampler lands within 12% of the 1000-step ancestral sampler at $5.2\times$ the throughput: the last 950 evaluations of the ancestral chain buy 1.4 FID points and cost 5.9 hours per 50k-sample evaluation. The 10-step sampler is the other end of the trade — $100\times$ fewer evaluations for $2.8\times$ the FID.

5.2 Quality–NFE frontier on MNIST

Table 6 is the same measurement for all four MNIST cells, which share a training budget and protocol. Two patterns stand out. First, the rectified-flow cells dominate ϵ -prediction once the budget exceeds ~ 16 evaluations: at 64 evaluations the flow cells reach FID 0.88 (UNet) and 4.99 (DiT), where the best ϵ -prediction result is 4.32 at 50. The crossover sits near NFE 16, where the two families are within 10% of each other. Second, every curve is monotone in the budget — with one exception, the transformer’s ϵ cell, which is the subject of the next subsection.

5.3 The deterministic sampler collapses on the transformer

Cell B is the one place in this study where more computation makes the result worse: its FID *grows* from 49.1 at NFE 10 to 159.7 at 20 and 672.4 at 50, while 1000-step ancestral sampling of the same weights gives 33.2. Four checks localize the cause.

Table 5: CelebA 64²: one trained model, four sampling budgets (10k samples each, EMA weights). Throughput is measured on the same runs; the last column projects the cost of a 50k-sample evaluation.

Sampler	NFE	FID	samples/s	50k evaluation
DDIM	10	31.40	49.5	17 min
DDIM	20	19.13	24.8	34 min
DDIM	50	12.75	9.9	1.4 h
ancestral	1000	11.37	1.9	7.3 h

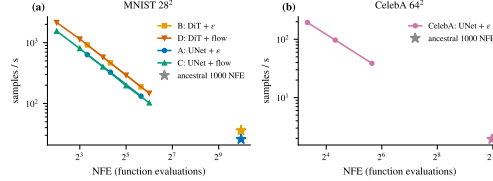


Figure 3: Sampling throughput against NFE, measured on the same runs. Throughput is inversely proportional to NFE (log–log slope ≈ -1 for every model), so the quality–NFE curves above are also quality–wall-clock curves up to a per-model constant.

Not the weights. Sampling the raw (non-EMA) weights reproduces the failure (FID 273.5 and 274.6 at NFE 20 and 50), so the exponential moving average is not at fault.

Not the model. The identical checkpoint with the stochastic ancestral sampler reaches 33.2, and the training-time sample grids (ancestral, EMA weights) are recognizable digits. The transformer learned the distribution well enough to be sampled.

Not the sampler code. The same function improves the UNet monotonically ($22.06 \rightarrow 4.57 \rightarrow 4.32$); its arithmetic does not depend on the backbone.

It is the missing noise injection. DDIM’s η interpolates between the deterministic update ($\eta = 0$) and the ancestral posterior variance ($\eta = 1$). At fixed NFE = 50, raising η restores the data statistics monotonically (Table 7) and turns the samples from saturated blobs back into digits.

Why this happens. The deterministic recursion integrates the model’s ϵ error coherently along the trajectory, whereas each ancestral step re-injects the schedule’s noise and thereby bounds the drift. The transformer’s error is large enough for that difference to matter: measured against the same noised batches (`eval/eps_error_by_t.py`), its ϵ MSE is 30–55% higher than the UNet’s across the whole trajectory, with the gap widest at low noise levels — precisely where the deterministic update reads its final image directly out of the model’s x_0 estimate. The same sampler that is the cheapest way to sample the UNet is the worst way to sample the transformer, and the transformer is the model whose per-sample cost is lowest.

The practical reading is not that DDIM is fragile in general — it is the best sampler in Table 5 and on the UNet — but that the NFE reduction it offers is conditional on score accuracy, and a controlled study is the only way to notice when the condition fails. A paper reporting only the UNet numbers would have shipped a sampler recommendation that inverts on the other backbone.

6 Backbone and objective (RQ1)

Table 8 is the headline comparison: four cells, one protocol, one dataset, and a FID for every one of them. The UNet wins both rows and both columns. Against the ϵ -objective it wins by $11.9\times$ (2.79 vs. 33.18); against rectified flow it wins by $5.7\times$ (0.88 vs. 4.99). The ordering holds at every sampling budget we measured (Table 6), so it is not an artefact of one sampler.

Two caveats belong next to that table, and they cut in opposite directions.

Parameters are not compute. The DiT has *more* parameters (16.5M vs. 11.7M) and *less* compute (1.08 vs. 2.43 GFLOPs per sample): its parameters sit in a wide MLP over 49 tokens, while the UNet spends its compute on convolutions at 28². The comparison is therefore not compute-matched, and

Table 6: MNIST FID (10k samples, EMA weights) as a function of the sampling budget. “—” marks budgets that were not run for that cell. The 1000 column is the ancestral DDPM sampler; the other columns are DDIM (for ϵ cells) or Euler (for flow cells).

Cell	Sampler	10	16	20	32	50	64	1000
A: UNet + ϵ	DDIM	22.06	—	4.57	—	4.32	—	2.79
B: DiT + ϵ	DDIM	49.14	—	159.68	—	672.39	—	33.18
C: UNet + flow	Euler	—	5.06	—	1.79	—	0.88	—
D: DiT + flow	Euler	—	12.98	—	7.03	—	4.99	—

Table 7: DiT + ϵ , DDIM at NFE = 50, as a function of the noise-injection level η . “Saturated” is the fraction of pixels pinned at ± 1 after the final clamp; MNIST digits are near-binary, so the ancestral sampler (NFE 1000, $\eta = 1$) saturates 84.8% of pixels. Nothing else about the sampler changes.

η	0	0.25	0.50	0.75	1.00
saturated pixels	47.7%	57.7%	66.0%	74.3%	79.1%
mean $ x $	0.904	0.908	0.916	0.927	0.934

the mismatch favours the winner — it would be a stronger claim if the two cells had equal FLOPs. This is the compute-matched experiment the report does not run.

The objective comparison is the cleaner of the two. Cells C and D differ from A and B only in the regression target and the sampler, and the same ordering appears at every budget: rectified flow reaches FID 0.88 where ϵ -prediction plateaus at 4.32. The crossover sits near NFE 16, so the flow objective buys a $8\times$ reduction in evaluations on the cheap end of the curve, and a $4.9\times$ lower FID on the expensive end. This is the expected result — a straighter path is easier to integrate — and it reproduces on both backbones, which is what makes it credible.

7 Systems (RQ3)

Mixed precision is the whole story. BF16 buys $3.15\times$ the step rate and $1.6\times$ less peak memory, and nothing else we tried moves the needle. That single number is what makes the larger-scale experiments in this project affordable at all: the CelebA cells below cost ~ 36 and ~ 55 GPU-hours in fp32 and under 12 and 18 in bf16.

Compilation buys nothing here. `torch.compile` returns $1.00\times$ — the same step time to three digits — which is a negative result worth stating plainly. Two explanations are consistent with the profile: 67% of GPU time is already inside cuBLAS matmuls that have little to fuse, and the fusible elementwise work sits between large kernels rather than accumulating in a long chain.

Why FSDP loses at this scale. FSDP is 12.8% slower than DDP and saves only 4.8% of peak memory — which is the prediction this benchmark was built to check. The peak is dominated by activations: at batch 160 the model’s parameters, gradients and Adam state together account for roughly 2 GB of the 33.5 GB peak, so sharding them cannot help, while the all-gather and reduce-scatter on the critical path cost real time. Fully sharded training is a tool for models whose *states* dominate their memory; at 130M parameters on 48 GB cards, that condition is not met.

Reading the MFU column. The 10.4% fp32 and 32.7% bf16 figures come from measured FLOPs (46.0 GFLOPs per sample, forward, counted by `torch.utils.flop_counter`) against the A6000’s dense peak. Two caveats apply: the counter covers matmul and convolution work only, so elementwise-heavy operations are invisible to it (a fused-adaLN kernel would have to earn its keep out of the 15.6% slice in Fig. 6, not out of the FLOPs number); and this is a 130M-parameter model at 256 tokens, so neither the batch nor the sequence length is large enough to saturate tensor cores. The honest summary is that bf16 moved this workload from “not using the tensor cores” to “using them moderately”.

8 What didn’t work

Each entry follows the same shape: hypothesis, setup, observation, diagnosis, resolution, lesson.

Table 8: The 2×2 design on MNIST, at the best budget measured for each cell (1000-step ancestral for the ϵ cells, 64-step Euler for the flow cells; both 10k samples, EMA weights). Sizes are measured, not configured, and by the same tool for both backbones.

Cell	Backbone + objective	Params	GFLOPs/sample	Best FID	NFE	s/step
A	UNet + ϵ	11.7M	2.43	2.79	1000	0.101
B	DiT + ϵ	16.5M	1.08	33.18	1000	0.073
C	UNet + flow	11.7M	2.43	0.88	64	0.102
D	DiT + flow	16.5M	1.08	4.99	64	0.072

Table 9: Training step cost for the CelebA DiT at 160 images per GPU under each execution setting (30 steps, 5 warmups discarded; single GPU unless noted). The last two rows are the distributed comparison at four GPUs. Throughput is global for the multi-GPU rows.

Setting	GPUs	s/step	img/s	peak GB	MFU	speedup
FP32	1	1.372	117	32.9	10.4%	1.00×
BF16	1	0.436	367	21.1	32.7%	3.15×
compile	1	1.377	116	32.9	10.4%	1.00×
BF16 + compile	1	0.439	365	21.1	32.5%	3.13×
FP32 + DDP	4	1.408	455	33.5	10.1%	—
FP32 + FSDP	4	1.588	403	31.9	9.0%	0.89× vs. DDP

Learning rate 5×10^{-4} collapses training. *Hypothesis:* the 5×10^{-4} rate used by the first CelebA run was safe because Adam is self-normalizing. *Observation:* the loss diverged and the samples degraded into noise; the run produced no usable checkpoint. *Diagnosis:* at that rate the early updates in the high-noise regime were large enough to leave the region where the ϵ -prediction loss is informative, and the optimizer could not return. *Resolution:* the protocol in this report uses 2×10^{-4} with gradient clipping at 1.0, and logs the gradient norm every step so the failure is visible before the loss is. *Lesson:* the learning rate and the clipping threshold are part of the protocol, not tuning details — which is why every cell here shares one pair of them.

DataParallel deadlocks with a multi-worker DataLoader. *Hypothesis:* `nn.DataParallel` is the cheap way to use several GPUs for a model that fits on one. *Observation:* the first multi-GPU configuration hung, reliably, with no error output, when the DataLoader used worker processes. *Diagnosis:* DataParallel replicates the module per forward pass and serialises it through a single process, which interacts badly with forked workers pinning memory. *Resolution:* the training stack was rewritten around `torchrun`-native DDP, which is what every run in this report uses. *Lesson:* for anything beyond a single forward pass, DDP is both simpler and faster.

P2P hangs on this node’s NUMA layout. *Hypothesis:* with DDP in place, the remaining hang was a software bug. *Observation:* `nvidia-smi topo -m` showed the six GPUs split across two NUMA domains (0–3 and 4–5) with the cross-domain link traversing PCIe and the inter-socket interconnect; a job spanning both domains hung in the first collective while a job confined to one domain ran. *Diagnosis:* NCCL’s peer-to-peer transport over that link does not complete. *Resolution:* `NCCL_P2P_DISABLE=1`, plus a rule that DDP jobs stay inside one domain — both of which every multi-GPU measurement in this report honours. *Lesson:* on shared hardware the topology is part of the experiment setup, and it is cheaper to measure it once than to debug it later.

FSDP is a net loss at this scale. *Hypothesis:* sharding parameters, gradients and optimizer state should reduce the memory peak and let the batch grow. *Observation:* at 130M parameters and batch 160 per GPU, FSDP ran 12.8% slower than DDP and saved 4.8% of peak memory (Table 9). *Diagnosis:* the peak is activation-dominated — the sharded states are ~ 2 GB of a 33.5 GB peak — so the all-gather pays for itself only when parameters rather than activations set the ceiling. *Resolution:* reported as a deliberate reverse benchmark rather than removed; the criterion it establishes (states $>$ activations) is the one to check before reaching for FSDP at this scale. *Lesson:* a negative result with a stated criterion is more useful than an unmeasured technique in the toolbox.

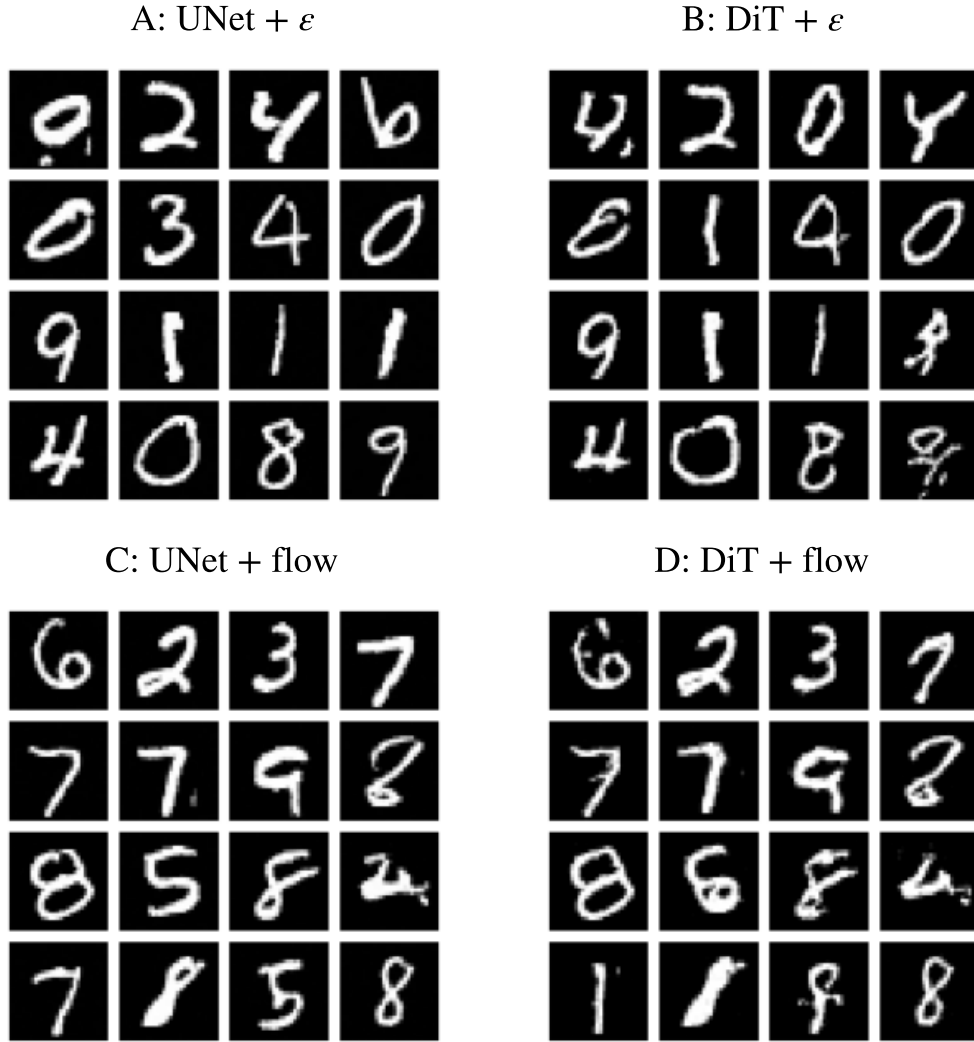


Figure 4: Final EMA samples, one block per cell (A: UNet + ϵ , B: DiT + ϵ , C: UNet + flow, D: DiT + flow). The transformer’s digits are thinner and more fragmented than the UNet’s, consistent with its FID.

9 Discussion and limitations

The three questions have different levels of support behind them. RQ2 is answered with a complete sweep at both scales. RQ3 is answered with measurements on one model and one hardware configuration, which is enough to establish the ranking of the levers (mixed precision $\gg$ everything else; compilation nothing; sharding negative) but not to generalise the constants. RQ1 has the weakest footing: the 2×2 is complete at MNIST scale, the comparison is not compute-matched, and there is no second scale — so the resulting ordering should be read as "at this size, under this protocol, with this much compute" rather than as a claim about transformers in general.

- Unconditional generation only: no classifier-free guidance, no text conditioning, no latent space.
- MNIST carries the full 2×2 ; CelebA contributes a sampler study on an existing checkpoint. The CelebA cells were estimated at ~ 36 h (DiT) and ~ 55 h (UNet) each in fp32 and were not run.

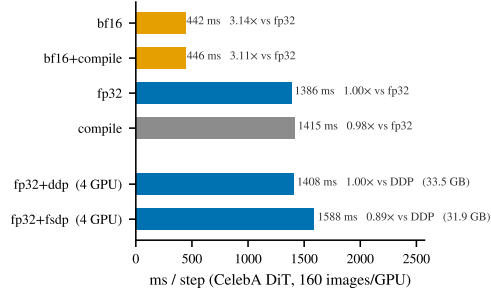


Figure 5: Step time by execution setting.

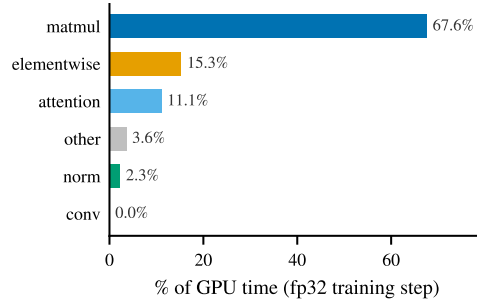


Figure 6: Where GPU time goes in a diffusion training step, by kernel category. This is a per-kernel accounting, which is precisely what a FLOPs count hides: the 15.6% spent on elementwise work does not appear in the 46.0 GFLOPs/sample figure at all.

- The A/B and C/D comparisons are not compute-matched: the UNet has $2.25\times$ the FLOPs per sample, so its win is achieved with more compute, not less.
- FID is computed against reference statistics built from this repository’s own preprocessing chain. Absolute values are *not* comparable with published numbers that use different resizing; only comparisons within this report are meaningful.
- The MNIST metric is a non-standard feature-space distance (a small CNN trained on MNIST for `eval/fid.py`), chosen because Inception features carry little information about grayscale digits.
- One seed per cell. The reported spread is FID variability across five random subsets, which is what the 10k-sample protocol would have given — not a confidence interval on the model.
- FLOPs counts cover matmul and convolution work only, so the MFU figures are a matmul-utilisation proxy rather than a hardware limit.
- The reported timings use fp32 training. bf16 was measured for *speed* (Table 9) but its effect on sample quality was not measured, so no quality number in this report comes from a mixed-precision run.
- The transformer’s deterministic-sampler collapse (Section 5) is characterised at one scale with one model. The η sweep localises the mechanism — the absence of noise injection — but where the boundary lies, and whether it moves with more training or more parameters, is not measured.

Reproduction

All commands (environment, data, smoke tests, launching cells, evaluation) are in `reports/runbook.md`; the raw archive behind every number is `results/*.json`; the committed code contains the full training stack and the test suite used to validate the implementations (`tests/test_sanity.py`, 14 CPU-only invariants).

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 2020.
- [2] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *ICLR*, 2021.
- [3] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *ICLR*, 2021.
- [4] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. *ICML*, 2021.
- [5] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv:2207.12598*, 2022.
- [6] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. *CVPR*, 2022.
- [7] William Peebles and Saining Xie. Scalable diffusion models with transformers. *ICCV*, 2023.
- [8] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: learning to generate and transfer data with rectified flow. *ICLR*, 2023.
- [9] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. *ICLR*, 2023.
- [10] Patrick Esser et al. Scaling rectified flow transformers for high-resolution image synthesis. *ICML*, 2024.
- [11] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *NeurIPS*, 2017.
- [12] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: fast and memory-efficient exact attention with IO-awareness. *NeurIPS*, 2022.
- [13] Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. ZeRO: memory optimizations toward training trillion parameter models. *SC20*, 2020.
- [14] Yanli Zhao, Andrew Gu, Rohan Varma, Liang Luo, Chien-Chin Huang, Min Xu, Less Wright, Hamid Shojanazeri, Myle Ott, Sam Shleifer, et al. PyTorch FSDP: experiences on scaling fully sharded data parallel. *VLDB*, 2023.